

JULIANO LORENZATO

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SUMMARY

I am a **Software Engineer** and **Computer Science Researcher** focused on computer graphics, GPU programming, and high-performance computing. My work explores the intersection of real-time rendering, heterogeneous computing, and programming systems for modern GPU architectures.

My current research focuses on surface detail representation techniques, particularly **relief texture mapping**. I investigate methods to simplify the implementation of relief-based rendering techniques and make them easier to integrate into real-time rendering pipelines. My goal is to reduce implementation complexity while preserving visual fidelity, enabling broader adoption of advanced shading techniques in graphics engines and GPU-driven applications.

In addition to my research work, I have professional experience building backend systems and developer tooling for web platforms, working on scalable APIs, distributed services, and infrastructure for production environments.

RECENT WORK EXPERIENCE

Scaler Integrated multiple third-party SSPs into a high-traffic web platform, implementing robust error-handling and automated retry flows using Inngest to ensure reliability, observability, and fault tolerance.	Nov/2025 - Present
Federal University of Pelotas I work as a researcher in the field of developing new languages for parallel and distributed programming, with a strong focus on heterogeneous environments. I have developed new ways to use the Elixir language for software development using GPUs and metaprogramming.	Aug/2023 - Present
Distu I improved and maintained a GraphQL API in Elixir/Phoenix/Absinthe using Oban and PostgreSQL for scheduling computationally intensive asynchronous tasks. Additionally, I developed an integration with a self-hosted AI model (Llama 3.1) for user support.	Jul/2024 - Nov/2025

SET OF HARD SKILLS

- C++, C, Rust, Go, Python, Lua, JavaScript/TypeScript.
- Unreal Engine, Godot Engine, Blender.
- WebGPU, WebGL2, Vulkan, OpenGL, WebAssembly, CUDA.
- Linear Algebra, Statistics, Calculus, Numerical Analysis, Discrete Mathematics, Physics.

EDUCATION

M.Sc. Computer Science • <i>Federal University of Rio Grande do Sul (UFRGS)</i>	2026-2028
B.Sc. Computer Science • <i>Federal University of Pelotas (UFPEl)</i>	2022-2026